

B1.4 Photosynthesis

Summary

Life processes depend on photosynthesis. Green plants and algae trap light from the Sun to fix carbon dioxide with hydrogen from water making organic compounds.

Underlying knowledge and understanding

Learners should also have some underpinning knowledge of photosynthesis. They should have an understanding that plants make carbohydrates in their leaves by photosynthesis, and be able to recall the word equation for photosynthesis.

Common misconceptions

Learners often think that plants do not respire.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM1.4i	understand and use simple compound measures such as the rate of a reaction	M1a and M1c
BM1.4ii	translate information between graphical and numerical form	M4a
BM1.4iii	plot and draw appropriate graphs, selecting appropriate scales and axes	M4a and M4c
BM1.4iv	extract and interpret information from graphs, charts and tables	M2c and M4a
BM1.4v	understand and use inverse proportion – the inverse square law and light intensity in the context of factors affecting photosynthesis	M1c

B1.4a	describe photosynthetic organisms as the main producers of food and therefore biomass for life on Earth				Use of concept cartoons to start discussions about photosynthesis.
B1.4b	describe the process of photosynthesis	reactants and products, two-stage process, location of the reaction (in the chloroplasts)		WS2a, WS2b, WS2c, WS2d	Investigation of photosynthesis e.g. the Priestley experiment using <i>Cabomba</i> to collect oxygen or the Ingenhousz experiment to show mass gain. (PAG B4, PAG B5, PAG B6)
B1.4c	describe photosynthesis as an endothermic reaction			WS1.3b, WS1.3c, WS1.3e	Demonstrate of an endothermic reaction (e.g. icepack).
B1.4d	describe experiments to investigate photosynthesis			WS2a, WS2b, WS2c, WS2d	Experiments to show the consequences of light exclusion on photosynthesising plants (e.g. testing geraniums for starch). (PAG B4, PAG B5, PAG B6)
B1.4e	explain the effect of temperature, light intensity and carbon dioxide concentration on the rate of photosynthesis		M1a, M1c, M4a, M4b, M4c, M2g	WS2a, WS2b, WS2c, WS2d	Investigation of photosynthesis in algae using alginate beads to immobilize the algae. (PAG B4, PAG B5, PAG B6)
B1.4f	explain the interaction of temperature, light intensity and carbon dioxide concentration in limiting the rate of photosynthesis	using graphs depicting the effects of the limiting factors	M1d, M2c, M4a, M1c	WS1.2b, WS1.2c, WS1.2e WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3f, WS1.3g, WS1.4e, WS2c, WS2d	